EIGRP

Course Code: CSC 3116 Course Title: Computer Networks

Dept. of Computer Science
Faculty of Science and Technology

Lecturer No:	Lab 6	**Week No:**		**Semester:**	Summer 22-23
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Lecture Outline

1. Introduction
2. Metric
3. Neighbor Discovery
4. EIGRP Tables
5. EIGRP Tables
6. EIGRP Configuration

Introduction

RIP vs EIGRP

	RIP	EIGRP
1	It support maximum 15 routers in the network. 16 router is unreachable.	It supports maximum 255 routers in the network. However, the default is 100 routers. (highly scalable)
2	Slow convergence	Fast convergence due to feasible successor
3	In RIP routing protocol, we cannot create a separate administrative boundary in the network.	In EIGRP routing protocol we can create a separate administrative boundary in the network with the help of autonomous system No. Less routing Table exchange is required.
4	It calculates the metric In terms of Hop Count from source network to destination network.	It calculates the metric In terms of bandwidth and delay (default).
5	RIP works on Bellman Ford algorithm.	EIGRP works on DUAL(Diffusing Update Algorithm) Algorithm.
6	It only maintains the best route to each destination.	It maintains the best route and some other alternative routes for each destination.
7	It is basically use for smaller size organization.	It is basically use for medium to lager size organization in the network [1].

Metric

- Combination of different factors
  - Bandwidth
  - Delay
  - Load
  - Reliability

Default

Metric....

Bandwidth

- No. of bits that can be sent over a link (kbps)
- Depends on interface type
- Use *bandwidth <1-10,000,000>* command to set bandwidth in kbps
- This is not real bandwidth; real bandwidth depends on clock rate
- The *bandwidth* command only influence route selection by routing protocol
- If no bandwidth is set, the default bandwidth of an interface is considered
- Calculated as the lowest bandwidth among all links in a route

Metric….

Table II Default bandwidth and delay

Interface	*Bandwdith*	*Delay (microseconds)*
Serial (T1)	1544 Kbps	20,000
Ethernet	10 Mbps	1000
Fast Ethernet	100 Mbps	100
Gigbit Ethernet	1000 Mbps	10
10 Gigbit Ethernet	10 Gbps	10

Metric….

Delay

- *Delay* is a measure of the time for a packet to reach its destination over a route (In theory)
- In practice, it is a constant set by the network engineer
- To set delay for an interface, use *delay* <value> command
- The value can be anything between 10 to 167,772,140 microseconds
- If it is not set, the default value (Table II) of each interface comes into effect
- Calculated as sum of delays in exit interfaces of all routers in a route.

Metric....

Metric calculation

$$Metric = \left[\frac{10^7}{least\ bandwidth} + delay_{total}\right] \times 256$$

Units

Bandwidth: kbps

Delay: Tens of microsecond

If the total delay is 30 seconds, $delay_{total}$=30/10=3

Metric....

Metric calculation

$$Metric = \left[\frac{10^7}{least\ bandwidth} + delay_{total}\right] \times 256$$

Route: 1-4-2-B

$Least\ BW = 56\ kbps$

$Total\ delay = 100/10 + 100/10 + 2000/10 = 220$

$Metric = \left[\frac{10^7}{56} + 220\right] \times 256 = 45770496^*$

Route: 1-3-2-B

$Least\ BW = 128\ kbps$

$Total\ delay = 100/10 + 100/10 + 1000/10 = 120$

$Metric = \left[\frac{10^7}{128} + 120\right] \times 256 = 20030720$

Perform rounding in every steps of calculation

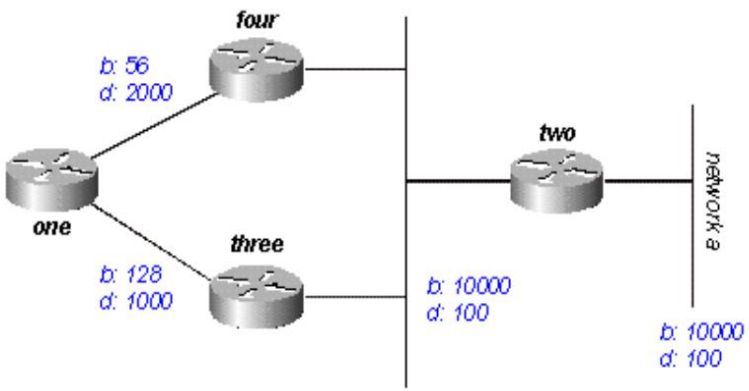


Fig. 1 A sample network [2]

Exercise

Calculate metric for all possible routes from router ONE to network A

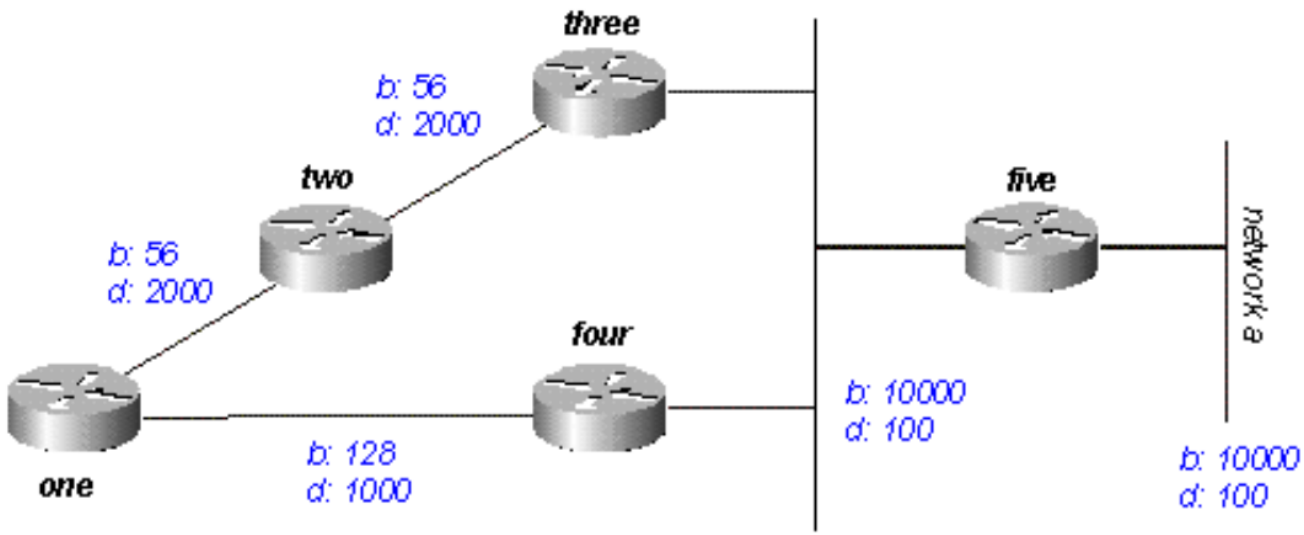


Fig. 2 A sample network [2]

Neighbor discovery

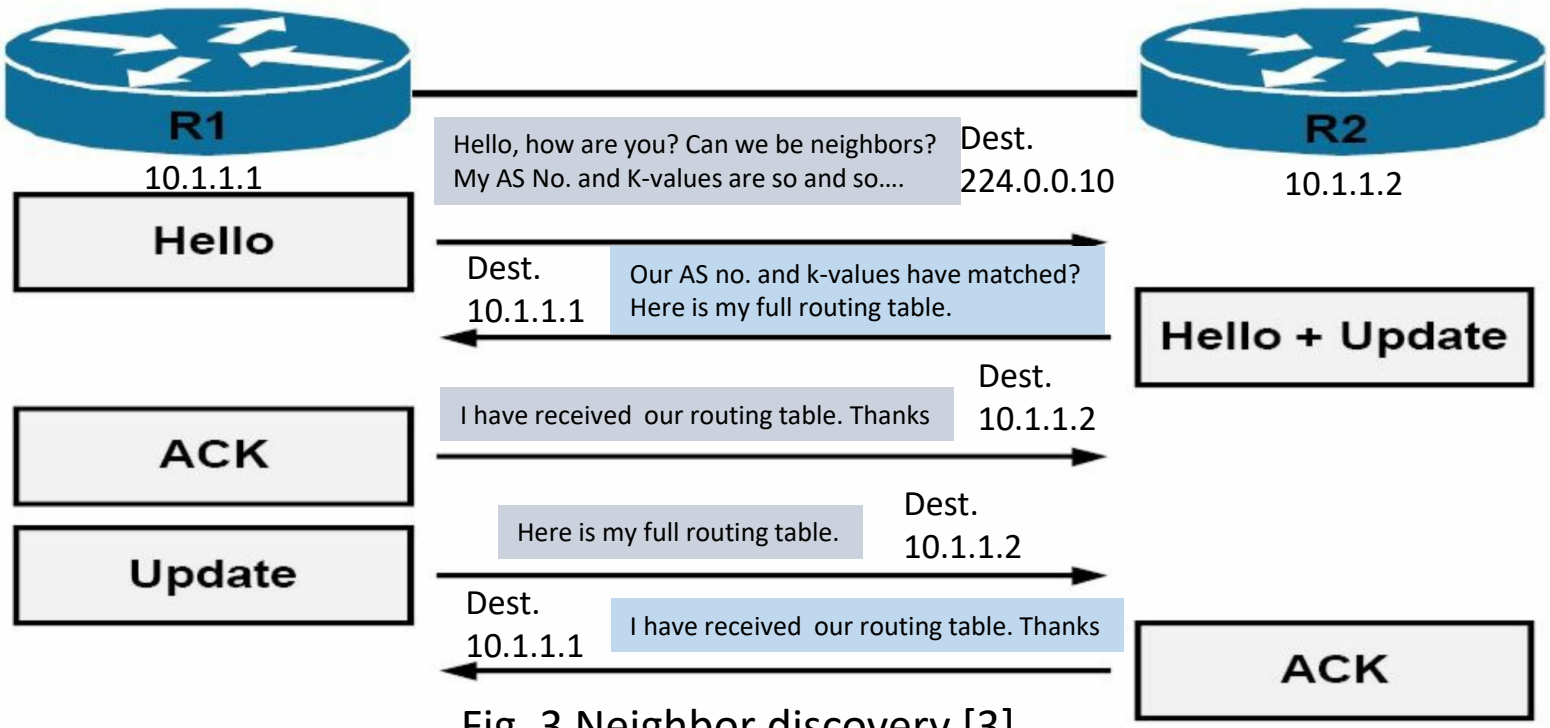


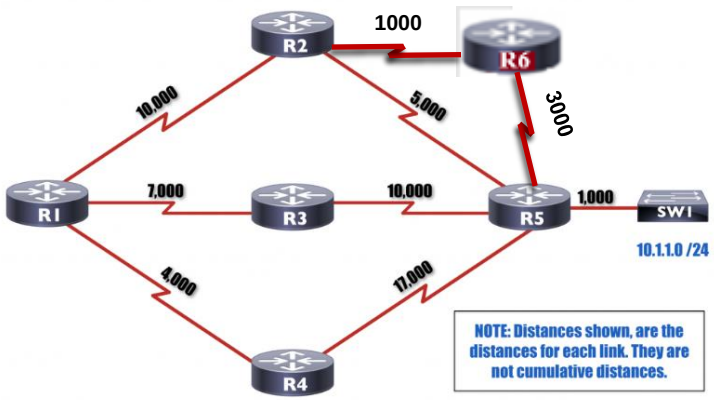
Fig. 3 Neighbor discovery [3]

Neighbor maintenance

- In neighbor discovery process, full routing table is sent
- Otherwise, only the change in routing table is sent
- After a neighbor discovery, the Hello packet is sent in every 5 second to know if the neighbor is still alive
- If a router does not receive any Hello packet from a neighbor within 15 seconds (called hold time), the neighbor is considered dead.
- For low-bandwidth link (e.g., T1), the periods are 60 sec and 180 sec.

EIGRP Tables....

Feasibility condition: reported distance must be less than the feasible distance through the successor



Neighbor	Reported Distance (RD)	Feasible Distance (FD)	(Feasible) Successor?
R2	6,000	16,000	Yes: Successor
R3	11,000	18,000	Yes: Feasible Successor
R4	18,000	22,000	No

Reported distance:
Distance advertised from neighbor as the distance between the Neighbor and the destination.

Feasible distance (FD):
Sum of Reported distance and distance between the router and the neighbor which reports the distance.

A is the route whose reported distance is less than the feasible distance of the best path.

EIGRP Tables

- Neighbor Table

  EIGRP shares routing information only with neighbors. To know who the neighbors are, it uses neighbor table. When a new neighbor is discovered, EIGRP would add its address and interface on which neighbor is connected in neighbor table [4].

- Topology Table

  Stores all feasible successors along with the successor (best route) for each destination network. EIGRP can store up to 32 feasible successors.

- Routing Table

  EIGRP stores single best route for each destination in this table. Router uses this table to forward the packet.

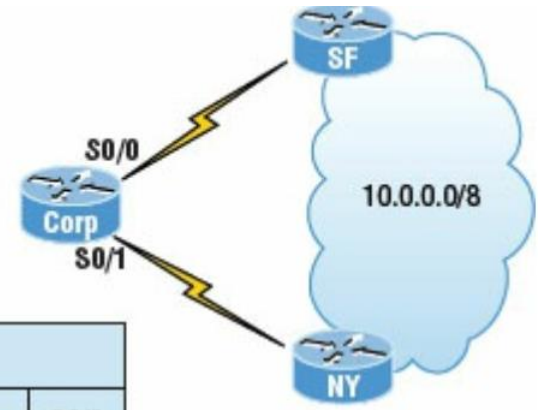
EIGRP Tables….

Example

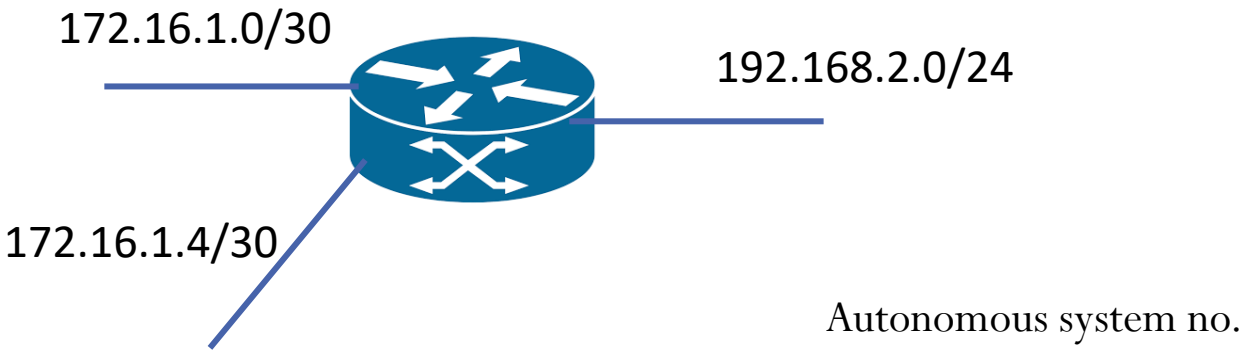
Neighbor Table—IP	
Router SF Router NY	Serial0/0 Serial0/1

Topology Table—IP			
SF NY	Destination 10.0.0.0 Destination 10.0.0.0	Successor Feasible Successor	S0/0 S0/1

Routing Table—IP	
SF Router/Exit S0/0	Successor



EIGRP Configuration



Autonomous system no.

RIP v2

R1(config)# router rip
R1(config-router)# version 2
R1(config-router)#network 172.16.1.0
R1(config-router)#network 172.16.1.4
R1(config-router)#network 192.168.2.0
R1(config-router)#no auto-summary

EIGRP

R1(config)# router eigrp <1 -65535>
R1(config-router)#network 172.16.1.0 0.0.0.3
R1(config-router)#network 172.16.1.4 0.0.0.3
R1(config-router)#network 192.168.2.0 0.0.0.255
R1(config-router)#no auto-summary

EIGRP Configuration….

- To show protocol properties
  show ip protocols
- To show neighbor table
  show ip eigrp neighbor
- To show topology table
  show ip eigrp topology
- To show routes made by eigrp
  show ip route eigrp
- To set maximum hops
  metric maximum-hops <1-255>
- To set the number of feasible successors
  maximum-paths <0-32>

References

[1] Computer Networking Class, https://computernetworkingclass.blogspot.com... /2016/08/comparison-between-rip-eigrp-igrp-and.html, [Accessed: April. 27, 2020].

[2] Cisco, “https://www.cisco.com/c/en/us/support/docs/ip/enhanced-interior... -gateway-routing-protocol-eigrp/16406-eigrp-toc.html, [Accessed: April. 27, 2020].

[3] P. Browning, F. Tafa, D. Gheorghe, and D. Barinic, *Cisco CCNA in 60 Days*, Reality Press Ltd., UK, 2014, pp. 581

[4] Computer Networking Notes, https://www.computernetworkingnotes.com/ccna... -study-guide/eigrp-tutorial-basic-concept-explained.html, [Accessed: April. 27, 2020].

Recommended Books

1. **Official Cert Guide CCNA 200-301 , vol. 1**, *W. Odom*, Cisco Press, First Edition, 2019, USA.
2. **CCNA Routing and Switching**, *T. Lammle*, John Wily & Sons, Second Edition, 2016, USA.